

Final Internship Report

Mechaline Internship

Vehicle Health AI and Driver Behaviour & Road Safety AI

1. Introduction

This report presents the work completed during my internship at **Mechaline**, covering the different stages of the internship from initial dataset development and computer vision work to driver behaviour analysis and machine learning.

The internship provided practical experience in artificial intelligence, computer vision, data analysis, machine learning, feature engineering, model evaluation, and technical documentation.

During the internship, my work primarily involved two related AI applications in the automotive and road-safety domain:

1. **Vehicle Health AI**, focused on detecting vehicle defects such as rust, corrosion, oil leaks, and coolant leaks using computer vision.
2. **Driver Behaviour & Road Safety AI**, focused on analyzing acceleration and braking behaviour and identifying potentially aggressive driving events.

The internship therefore provided experience with both **image-based AI** and **sensor-data-based machine learning**.

2. Internship Objectives

The overall objectives of the internship were to gain practical experience in AI development and contribute to automotive-focused AI projects.

The major objectives included:

- Collecting and preparing real-world datasets.
- Performing data cleaning and preprocessing.
- Annotating images for object detection.
- Developing and evaluating a YOLOv11 object detection model.
- Conducting exploratory data analysis on vehicle sensor data.
- Engineering useful acceleration and braking features.
- Developing a machine-learning model for aggressive driving detection.
- Evaluating model performance using appropriate metrics.

- Performing false-positive and false-negative analysis.
 - Testing models on unseen data.
 - Developing a prototype Driver Safety Score framework.
 - Preparing technical documentation and project handover materials.
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3. Week 1 — Vehicle Defect Dataset Collection

The first stage of the internship focused on developing a dataset for the **Vehicle Health AI** project.

The objective was to collect high-quality real-world images of vehicle defects that could later be used for object detection using the YOLOv11 framework. The initial focus was on defects such as rust, corrosion, and oil leaks.

Images were collected from automotive maintenance websites, repair blogs, inspection reports, and image-search platforms. The images were organized into separate folders according to defect category.

Preference was given to clear, high-resolution photographs taken in real workshop or inspection environments rather than diagrams or synthetic illustrations. Image sources were also documented to maintain traceability.

One of the major challenges was obtaining sufficient high-quality oil-leak images. Rust and corrosion images were comparatively easier to find, while oil-leak images required searching through multiple automotive forums and maintenance websites.

Another challenge was avoiding duplicate images and ensuring that images were suitable for later object detection. Images with poor lighting, excessive background clutter, low resolution, or unclear defects were removed.

By the end of the week, a structured image dataset had been established as the foundation for the annotation and model-development stages.

4. Week 2 — Image Annotation and Dataset Preparation

The second stage focused on converting the collected images into a dataset suitable for YOLOv11 object detection.

The images were uploaded to **Roboflow**, where each image was manually reviewed and annotated using the Object Detection annotation tool.

Four defect classes were used:

- Rust
- Corrosion
- Oil Leak
- Coolant Leak

Tight bounding boxes were drawn around visible defects. When multiple defects of the same class were present, separate bounding boxes were created.

Images that were blurry, duplicated, or did not clearly show the assigned defect were excluded.

The dataset was then preprocessed by correcting image orientation and resizing images to **640 × 640 pixels**.

Data augmentation was applied to the training data, including horizontal flipping and slight image rotation. The validation and test sets were kept unchanged so that model evaluation would remain unbiased.

YOLO-compatible label files were generated containing class identifiers and normalized bounding-box coordinates.

A major challenge during annotation was distinguishing **rust from corrosion** because they can look visually similar. Consistent definitions were therefore established for the two categories. Another challenge involved differentiating oil leaks from coolant leaks when the leaked liquid was not clearly identifiable.

Ambiguous images were excluded whenever the correct class could not be determined reliably.

5. Week 3 — Vehicle Health AI Model Development

The third stage involved training and evaluating the Vehicle Health AI object-detection model.

A **YOLOv11s** object detection model was trained to identify the four vehicle-defect classes:

- Coolant Leak
- Corrosion
- Oil Leak
- Rust

The training was performed using **Google Colab and a Tesla T4 GPU** for 50 epochs. The dataset consisted of manually annotated images prepared using Roboflow and resized to 640 × 640 pixels.

The overall validation results were:

- Precision: **43.2%**
- Recall: **33.1%**

- mAP@50: **33.9%**
- mAP@50-95: **18.4%**

The class-wise performance showed that coolant leak had the strongest mAP@50 performance, while corrosion and rust were more difficult for the model to detect reliably.

The model was also tested on **20 unseen vehicle images** that were not part of the training or validation datasets.

The model successfully detected several vehicle defects, although false negatives were also observed. Two notable false positives occurred when the model incorrectly associated green objects with coolant leaks.

This indicated that the model had learned some visual characteristics, such as green coloration, that were not necessarily unique to coolant leaks.

Based on the analysis, recommendations included increasing the number of annotated images, especially for rust and corrosion, adding coolant leaks with different colors, and including negative examples such as green vehicles, vegetation, and other green objects.

The YOLOv11 model therefore provided a functional baseline for future development of the Vehicle Health AI system.

6. Transition to Driver Behaviour & Road Safety AI

Following the vehicle-defect detection work, the internship progressed toward **Driver Behaviour & Road Safety AI**.

The objective of this work was to investigate whether vehicle and smartphone sensor data could be used to identify unsafe driving behaviour.

The project focused particularly on acceleration and braking behaviour.

The research considered the use of accelerometers, gyroscopes, GPS, vehicle speed, braking information, steering behaviour, lane changes, and other sensor information for driver-behaviour analysis.

Three publicly available datasets were reviewed during the research phase, including a Harsh Driving Dataset, a smartphone sensor dataset for driver behaviour analysis, and the UAH-DriveSet.

The research showed that smartphone sensors can provide a relatively low-cost method for monitoring driving behaviour because modern smartphones contain accelerometers, gyroscopes, GPS and other sensors.

Potential applications identified during the research included:

- Driver Safety Score systems
 - Driver behaviour classification
 - Fleet management
 - Driver coaching
 - Insurance telematics
 - Accident-risk prediction
 - Road-safety monitoring
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7. Acceleration and Braking Data Analysis

The next stage focused specifically on acceleration and braking behaviour.

The selected dataset contained **16,388 observations** and included accelerometer, gyroscope, magnetometer, road-type and driving-event information.

Six driving-event categories were analyzed, including sudden acceleration, sudden braking, uniform acceleration and constant acceleration conditions.

An acceleration-magnitude feature was calculated using the three accelerometer axes.

The analysis showed that the acceleration-magnitude distribution was right-skewed, meaning that most observations had relatively low-to-moderate acceleration magnitudes while a smaller number contained much larger values.

The mean acceleration magnitude for the main event categories was:

- Sudden acceleration: **5.04**
- Sudden braking: **4.78**
- Uniform acceleration: **4.40**
- Sudden acceleration alternative: **4.23**
- Constant acceleration at 40 km/h: **3.85**
- Constant acceleration at 20 km/h: **3.53**

These results indicated that sudden acceleration and sudden braking generally produced stronger acceleration behaviour than controlled acceleration events.

8. Outlier Analysis

Outlier analysis was performed using the **Interquartile Range (IQR)** method.

The calculated values were approximately:

- Q1 = **2.5967**
- Q3 = **5.6899**
- IQR = **3.0931**
- Upper outlier boundary = **10.3296**

Observations above the upper boundary were considered statistical outliers.

However, these observations were not automatically removed. In this project, very high acceleration values could represent genuine harsh-driving events rather than measurement errors.

The analysis showed that sudden acceleration and sudden braking contained more extreme observations than controlled acceleration conditions.

This supported the idea that acceleration magnitude and acceleration variability could be useful indicators for distinguishing aggressive from smoother driving behaviour.

9. Week 6 — Driver Behaviour Statistical Analysis

The larger driver-behaviour dataset contained **32,987 records**.

The dataset contained:

- 4,698 harsh-braking records
- 11,690 acceleration records
- 483 safe/no-movement records
- 16,116 other records

The mean acceleration magnitude across the dataset was approximately **4.42**, with a median of approximately **3.759** and a maximum of **69.92**.

The behaviour-group statistics showed:

- Acceleration: mean 4.523
- Harsh braking: mean 4.782
- Safe/no-movement: mean 1.563

The considerably lower mean acceleration magnitude for safe/no-movement observations provided additional evidence that acceleration intensity could help characterize different driving behaviours.

10. Feature Engineering

Feature engineering was performed to transform the raw sensor measurements into features that could better represent driving behaviour.

The final machine-learning feature set contained 13 features:

- `acc_x`
- `acc_y`
- `acc_z`
- `acc_magnitude`
- `horizontal_acc`
- `vertical_acc`
- `acc_x_abs`
- `acc_y_abs`
- `acc_z_abs`
- `acc_mean_5`
- `acc_std_5`
- `acc_change`
- `acc_change_abs`

The features captured several different aspects of vehicle motion.

The raw acceleration features represented the three acceleration axes.

Acceleration magnitude represented the overall intensity of acceleration.

Horizontal and vertical acceleration represented different components of the motion.

Absolute acceleration features captured the magnitude of acceleration without considering direction.

Rolling features such as `acc_mean_5` and `acc_std_5` captured short-term acceleration behaviour and variability.

The acceleration-change features represented transitions in acceleration between observations.

These engineered features provided the input for the machine-learning classification model.

11. Machine Learning Model

A **Random Forest classifier** was selected for classifying driving observations into:

- `0` — Non-Aggressive

- **1** — Aggressive

The model was trained using the engineered acceleration and braking features.

Feature-importance analysis showed that `acc_mean_5` was the most important feature, followed by `acc_std_5` and horizontal acceleration.

This indicates that short-term acceleration behaviour and variability were particularly useful to the model.

The model did not rely on a single acceleration threshold. Instead, it combined acceleration magnitude, direction, absolute values, short-term statistics and acceleration changes.

12. Threshold Optimization

The initial model showed that the default classification threshold was not suitable for the project's main objective.

Because the project aimed to detect as many potentially aggressive events as possible, different probability thresholds were evaluated.

Lowering the threshold increased recall while reducing precision.

A threshold of **0.15** was selected for the final held-out evaluation because it prioritized the detection of aggressive events.

This resulted in a high-sensitivity model that could detect a large proportion of aggressive events, although it also generated a relatively high number of false alarms.

13. Final Model Evaluation

The final Random Forest evaluation at the selected threshold of 0.15 produced:

- Accuracy: **45.51%**
- Precision: **26.83%**
- Recall: **86.20%**
- F1 Score: **40.93%**
- ROC-AUC: **64.75%**
- PR-AUC: **32.30%**

The confusion matrix contained:

- True Negatives: **899**
- False Positives: **1,737**

- False Negatives: **102**
- True Positives: **637**

The model detected **86.20% of aggressive observations**, which was valuable for a high-sensitivity safety-warning prototype.

However, the large number of false positives resulted in relatively low precision and accuracy.

Therefore, the model should be interpreted as a prototype designed to prioritize aggressive-event detection rather than as a production-ready driver safety classifier.

14. Error Analysis

False positives and false negatives were analyzed to understand where the model struggled.

The analysis showed that false positives generally had stronger acceleration characteristics than true negatives.

For example, the average acceleration magnitude was approximately:

- True Negative: 3.522
- False Positive: 5.285
- False Negative: 4.333
- True Positive: 6.031

The average acceleration standard deviation was also highest for true positives.

These results showed that some non-aggressive observations naturally resembled aggressive events in terms of acceleration intensity.

False negatives also demonstrated that aggressive driving cannot be identified using acceleration magnitude alone.

This supported the need for combining magnitude, direction, short-term variability and acceleration changes rather than relying on a single feature.

15. Genuine Unseen-Data Testing

A complete CSV driving session was excluded from model training and reserved for genuine unseen-data testing.

The selected unseen session was:

paved_sudden-right-line-chg_const-acc-at-20kmph_22-3-2023_19-37-54.csv

After feature preparation, the unseen session contained **470 usable observations**:

- 249 Non-Aggressive
- 221 Aggressive

The model was trained using the other 53 CSV sessions.

At a threshold of 0.20, the unseen-data results were:

- Accuracy: **53.62%**
- Precision: **50.41%**
- Recall: **83.26%**
- F1 Score: **62.80%**
- ROC-AUC: **61.86%**

The confusion matrix was:

- True Negative: 68
- False Positive: 181
- False Negative: 37
- True Positive: 184

The model therefore correctly identified **184 of the 221 aggressive observations** in the genuinely unseen session.

This provided evidence that the model retained useful aggressive-event detection capability on data that was not used for training. However, the false-positive rate remained a significant limitation.

16. Representative Unseen Predictions

Additional analysis was performed on a genuinely unseen recording containing **1,235 samples**.

At the selected threshold, the model produced:

- Predicted Non-Aggressive: 511
- Predicted Aggressive: 724
- Predicted Aggressive rate: 58.62%
- Mean aggressive probability: 0.292

The strongest aggressive predictions had probabilities of approximately:

- 0.865
- 0.860
- 0.840

- 0.825
- 0.820

The predicted aggressive percentage should be interpreted as a **model prediction rate**, rather than a direct measurement that 58.62% of the driver's behaviour was objectively unsafe.

17. Driver Safety Score Framework

The final stage involved developing a prototype Driver/Trip Safety Score.

The purpose of the score was to move from individual sensor observations toward a higher-level representation of driver behaviour over a trip.

The proposed score combines behavioural indicators such as:

- Predicted aggressive-event rate
- Harsh-braking rate
- Rapid-acceleration rate
- Acceleration variability
- Acceleration-change behaviour
- Driving smoothness

The proposed score ranges from 0 to 100.

The proposed interpretation is:

80–100 — Safe

Low behavioural risk.

60–79 — Moderate

Some risk indicators are present.

40–59 — Risky

Frequent risk indicators.

0–39 — High Risk

Strong behavioural risk indicators.

This Safety Score is a project-developed prototype and has not been externally validated against accident outcomes or an established industry safety-score standard.

18. Overall Internship Outcomes

The internship resulted in practical experience across several stages of an AI development lifecycle.

The major outcomes included:

Vehicle Health AI

A complete first-version object-detection pipeline was developed:

Raw images → Dataset collection → Annotation → Preprocessing → Augmentation → YOLOv11 training → Validation → Unseen-image testing → Error analysis.

The resulting YOLOv11s model achieved 43.2% precision, 33.1% recall, 33.9% mAP@50 and 18.4% mAP@50–95.

Driver Behaviour & Road Safety AI

A second AI pipeline was developed:

Raw sensor data → Cleaning → EDA → Feature engineering → Random Forest → Threshold optimization → Error analysis → Unseen-data validation → Safety Score framework.

The final project dataset contained 54 unique CSV files and approximately 32,987 raw records.

19. Challenges Encountered

Several technical challenges were encountered during the internship.

During the Vehicle Health AI phase, obtaining sufficient high-quality images for certain defect classes was difficult. Annotation also required careful distinction between visually similar defects such as rust and corrosion.

During model development, false positives and false negatives had to be investigated to understand why the model failed on certain images.

In the Driver Behaviour project, a major challenge was distinguishing aggressive behaviour from normal driving events with similar acceleration characteristics.

Another challenge was the trade-off between precision and recall. Lowering the classification threshold improved aggressive-event recall but also produced many false positives.

The unseen-data evaluation further demonstrated that model generalization remains challenging when the available data represents specific driving scenarios.

20. Skills Developed

The internship provided practical experience in:

- Python programming
- Pandas
- NumPy
- Matplotlib
- Scikit-learn
- Google Colab
- YOLOv11
- Roboflow
- Computer vision
- Object detection
- Image annotation
- Dataset preprocessing
- Data augmentation
- Exploratory data analysis
- Statistical analysis
- Feature engineering
- Random Forest classification
- Model evaluation
- Threshold optimization
- Confusion-matrix analysis
- Error analysis
- Unseen-data validation
- Technical documentation
- AI project presentation

The internship also strengthened my understanding of how machine-learning projects progress from raw data collection to model development, evaluation and final handover.

21. Limitations and Future Improvements

The Vehicle Health AI model can be improved by expanding the dataset, particularly for rust and corrosion, and by adding greater variation in lighting, viewpoints and defect appearance.

For the Driver Behaviour system, future improvements should include collecting data from more drivers and vehicles and increasing variation in road and traffic conditions.

Additional vehicle speed and speed-change information could also improve the analysis.

Longer temporal windows and stronger trip-level aggregation could provide a better representation of driving behaviour.

Other machine-learning approaches such as XGBoost, LightGBM and sequence-based models could be compared with the Random Forest baseline.

Driver-level train/test splitting should also be considered to provide a stronger estimate of generalization to completely new drivers.

Finally, the Driver Safety Score should be calibrated and validated against independently labelled driving sessions before being considered for real-world deployment.

22. Conclusion

The internship at Mechaline provided an opportunity to work on practical artificial-intelligence applications within the automotive and road-safety domain.

The internship began with the development of a vehicle-defect image dataset and progressed through manual annotation, preprocessing, YOLOv11 object detection, model evaluation and unseen-image testing.

The later phase focused on Driver Behaviour & Road Safety AI. This work involved researching driver-behaviour datasets, analyzing acceleration and braking patterns, performing exploratory data analysis, engineering behavioural features, training a Random Forest classifier, optimizing its classification threshold, analyzing model errors, and testing the model on genuinely unseen driving data.

The final Driver Behaviour system demonstrated strong sensitivity toward aggressive events, achieving 86.20% recall on the final held-out evaluation and 83.26% recall on the dedicated unseen session. At the same time, the high false-positive rate demonstrated the limitations of the current prototype.

Overall, the internship helped develop practical skills in computer vision, data analysis and machine learning while providing experience in the complete AI development process. The work also established foundations for future development of automated vehicle-health monitoring and driver-safety systems.